5.2.2 Balancing exploration and exploitation

In this subsection, we further talk about how the UCB policy may be adjusted by us the BayesOpt user, which offers a level of control that balances between regions with high uncertainty (exploration) and regions with high predictive mean (exploitation). This discussion will give us a deeper understanding of UCB, before moving ahead with BoTorch implementation in the next subsection.

Remember that with a normal distribution, going two standard deviations away from the mean (that is, mean μ plus/minus 2 times standard deviation σ) gives us the 95% credible interval (CI). The upper bound of this interval ($\mu + 2\sigma$) is the acquisition score of UCB we saw in the previous subsection.

However, the 95% CI is not the only CI of a normal distribution. By setting the multiplier, denoted as β, for the standard deviation σ in the formula $\mu + \beta\sigma$, we obtain other CIs.

For example, in a one-dimensional normal distribution as shown in Figure 5.7:

- Going one standard deviation above the mean (+), that is, setting = *1*, gives the 68% CI: 68% of the probability mass of the normal distribution lies between - and + .
- Similarly, three standard deviations away from the mean (= *3*) gives us the 99.7% CI.

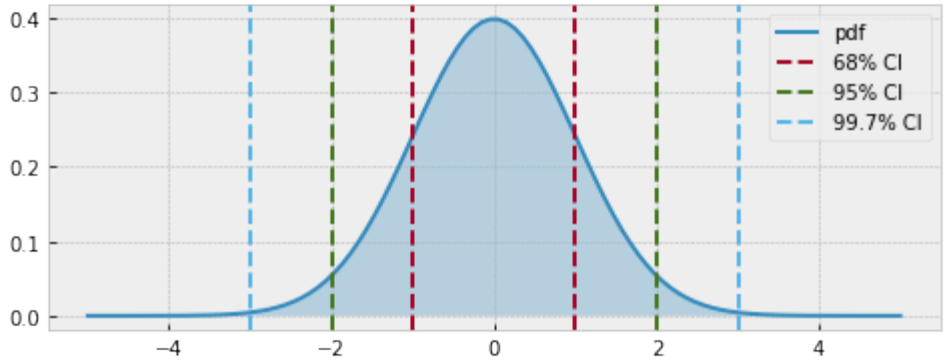


Figure 5.7 Different credible intervals of a standard normal distributions. Going away from the mean to one, two, and three standard deviations, we obtain the 68%, 95%, and 99.7% credible intervals. The upper bounds of these intervals are used by the Upper Confidence Bound policy.

In fact, any value of β would give us a unique CI for the normal distribution. Since UCB only dictates that we should use an upper bound from our predictive model to make decisions, any value of the form $\mu + \beta\sigma$ can serve as the upper bound used in UCB. By setting this parameter β, we can control for the behavior of the UCB policy.

Figure 5.8 shows three different Upper Confidence Bounds, corresponding to $\beta = 1, 2, 3$, while the mean function in fact corresponds to setting $\beta = 0$. We see that although the upper bounds are

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